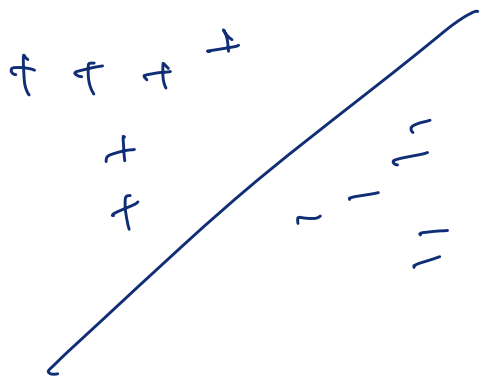


Generative learning also

Discriminative

$y|x, \theta \sim \text{Exponential}(\eta)$

$$\eta \leftarrow \theta^T x$$



Generative

$x|y \sim \mathcal{N}$

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

$$P(y) = \sum_{x'} P(x|y) P(x')$$



MGD | MVG

$$P(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0)\right)$$

$$E(x) = \mu$$

$$\begin{aligned} \text{Cov}(x) &= E[(x - E(x))(x - E(x))^T] \\ &= E[xx^T] - (E(x) E(x))^T \end{aligned}$$

Gaussian Discriminant Analysis

$$x|y=0 \sim \mathcal{N}(\mu_0, \Sigma)$$

$$x|y=1 \sim \mathcal{N}(\mu_1, \Sigma)$$

$$y \sim \text{Bernoulli}(\phi)$$

$$\checkmark p(y) = \phi^y (1-\phi)^{(1-y)}$$

$$p(x|y=0) =$$

$$\begin{aligned} \mathcal{L}(\Phi, \mu_0, \mu_1, \Sigma) &= \prod_{i=1}^n p(x^i, y^i, \Phi, \mu_0, \mu_1, \Sigma) \\ &= \prod_{i=1}^n p(x^i | y^i; \mu_0, \mu_1, \Sigma) p(y^i; \Phi) \\ &= \prod_{i=1}^n p(x^i | y^i=0; \mu_0, \Sigma) \\ &\quad p(x^i | y^i=1; \mu_1, \Sigma) p(y^i, \Phi) \end{aligned}$$

$$\begin{array}{l|l} \nabla_{\Phi} \mathcal{L} = 0 & \Phi = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{y^i=1\} \checkmark \\ \nabla_{\mu_0} \mathcal{L} = 0 & \mu_1 = \frac{\sum_{i=1}^n \mathbb{1}\{y^i=1\} x^i}{\sum_{i=1}^n \mathbb{1}\{y^i=1\}} \\ \nabla_{\mu_1} \mathcal{L} = 0 & \\ \nabla_{\Sigma} \mathcal{L} = 0 & \end{array}$$

$$\mu_0 =$$

$$\begin{aligned} \Sigma &= \frac{1}{n} \sum_{i=1}^n (x^i - \mu_{y(i)}) (x^i - \mu_{y(i)})^T \\ &= \frac{1}{n} \left(\sum_{i \in -} (x^i - \mu_0) (x^i - \mu_0)^T + \sum_{i \in +} (x^i - \mu_1) (x^i - \mu_1)^T \right) \end{aligned}$$

01/09/25

$$P(x|y=c; \theta) = N(x|M_c, \Sigma_c) \quad \pi_c = P(y=c)$$

$$P(y=c|x; \theta) \propto \underbrace{P(y=c)}_{\pi_c} \cdot N(x|M_c, \Sigma_c)$$

$$\log P(y=c|x; \theta) = \log \pi_c - \frac{1}{2} \log |2\pi\Sigma_c| - \frac{1}{2} (x-M_c)^T \Sigma_c^{-1} (x-M_c) + \text{Const}$$

$$= \log \pi_c - \frac{1}{2} x^T \Sigma_c^{-1} x + x^T \Sigma_c^{-1} M_c - \frac{1}{2} M_c^T \Sigma_c^{-1} M_c + \text{Const}$$

Quadratic discriminant Analysis (QDA)

let $\Sigma_c = \Sigma$

$$\log P(y=c|x; \theta) = \log \pi_c - \frac{1}{2} (x-M_c)^T \Sigma^{-1} (x-M_c) + \text{Const}$$

$$= \log \pi_c - \frac{1}{2} M_c^T \Sigma^{-1} M_c + \underbrace{x^T \Sigma^{-1} M_c}_{\beta_c} + \underbrace{\text{Const}}_K - \frac{1}{2} x^T \Sigma^{-1} x$$

Linear discriminant Analysis

$$\log P(y=c|x; \theta) = \gamma_c + x^T \beta_c + K$$

$$\Rightarrow \underline{P(y=c|x; \theta)} = \frac{e^{(\gamma_c + x^T \beta_c)}}{\sum_{c'} e^{(\gamma_{c'} + x^T \beta_{c'})}}$$

$$= \frac{e^{w_c^T [1, x]}}{\sum_{c'} e^{w_{c'}^T [1, x]}}$$

$$\underline{w_c = [\gamma_c, \beta_c^T]^T}$$

$$= \frac{e^{\beta_c^T x + \gamma_c}}{(\quad)}$$

$$= \frac{e^{\theta_c^T x}}{\sum_{c'} e^{\theta_{c'}^T x}}$$

multinomial Logistic Regress.

We maximized the conditional likelihood $P(y|w, c)$
 In LDA, we first fit Gaussian and class prior ($P(y=c)$)
 and then maximize the joint likelihood $P(x, y|\theta)$

Let's assume binary classification problem —

$$c=0; c=1$$

$$P(y=1|x; \theta) = \frac{e^{\beta_1^T x + \gamma_1}}{e^{\beta_1^T x + \gamma_1} + e^{\beta_0^T x + \gamma_0}} = \frac{1}{1 + e^{(\beta_0 - \beta_1)^T x + (\gamma_0 - \gamma_1)}}$$

$$\theta \sim \mathcal{N}(-((\beta_0 - \beta_1)^T x + (\gamma_0 - \gamma_1)))$$

$$= \mathcal{N}(\underline{(\beta_1 - \beta_0)^T x + (\gamma_1 - \gamma_0)})$$

$$\begin{aligned} \gamma_1 - \gamma_0 &= -\frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \frac{1}{2} \mu_0^T \Sigma^{-1} \mu_0 + \log(\pi_1/\pi_0) \\ &= -\frac{1}{2} (\mu_1 - \mu_0)^T \Sigma^{-1} (\mu_1 + \mu_0) + \log(\pi_1/\pi_0) \end{aligned}$$

$$w = \beta_1 - \beta_0 = \Sigma^{-1} (\mu_1 - \mu_0)$$

assume that $x_0 = \frac{1}{2} (\mu_1 + \mu_0) - (\mu_1 - \mu_0) \frac{\log(\pi_1/\pi_0)}{(\mu_1 - \mu_0)^T \Sigma^{-1} (\mu_1 - \mu_0)}$

$$w^T x_0 = -(\gamma_1 - \gamma_0)$$

$$\mathcal{N}(w^T x - w^T x_0)$$

$$P(y=1|x;\theta) = \mathcal{N}(w^T (x - x_0))$$

$$w^T x = \underbrace{w^T x_0}_c$$

$$\theta^T x = c$$

let's assume that $\pi_1 = \pi_0 = 0.5 \rightarrow c = w^T x_0 = \frac{1}{2} w^T (\mu_1 + \mu_0)$

if $\pi_1 = \pi_0$ then $x_0 = \frac{1}{2} (\mu_1 + \mu_0) \Rightarrow$ the halfway

if $\pi_1 > \pi_0$, we have to be closer to μ_0 than the halfway in order to pick class 0

$\Rightarrow$ we show that

LDA with multi class $\equiv$ multinomial Logistic Reg.

$\Downarrow$
LDA with 2 classes $\equiv$ Logistic Reg.

$$\sum_c \neq \sum \quad \sum_c \quad x \in \mathbb{R}^d$$

if $|c| = \text{no of classes} \ll d \Rightarrow \text{overfit}$

Assume that all features are independent
and we need $\mathcal{O}(|c|d)$ parameters

$\Downarrow$
Naive Bayes Algo

$$P(y|x) = \frac{P(x|y) P(y)}{P(x)}$$

$$= P(\langle x_1, x_2, \dots, x_d \rangle | y) P(y)$$

$$= [P(x_1|y) \cdot P(x_2|y) \cdot \dots \cdot P(x_d|y)] \underline{P(y)}$$

Spam Non-Spam

$$P(\text{dollar} | \text{Spam})$$

$$P(\text{dollar} | \text{non-Spam}) \quad P(\text{Spam}) \approx \frac{50}{90}$$

$$\frac{1}{500}$$

$$\left. \begin{array}{l} \text{dollar} \\ \text{reward} \\ \text{cash} \end{array} \right\} 50 \quad \left. \begin{array}{l} \text{ } \\ \text{ } \\ \text{ } \end{array} \right\} 40 = 10/500$$

Smoothing

Add-one smoothing

$$P(\text{dollar} | \text{Spam})$$

$$= \Rightarrow \frac{\text{Count}(\text{dollar}, \text{Spam}) + 1}{\text{Count}(\# \text{ words in Spam})}$$

$$P(\text{reward} | \text{Spam}) = \frac{C(\text{reward}) + 1}{(\quad) + 1}$$

$$P(\text{cash} | \text{Spam}) = \frac{C(\text{cash}) + 1}{(\quad) + d} \quad \text{d.d}$$

$$\text{Vocabulary} = V$$

$$d$$